# MINI PROJECT REPORT

On

# Bl-Based Job Market Demand Analysis Dashboard

By

Shreyas Sadavarte (BECOMPC47)

Under the guidance of

Prof. Nitanjali Mane

Department of Computer Engineering
Pimpri Chinchwad College of Engineering and Research, Ravet Plot No: B, Sector No. 110, gate no 1, Laxmi Nagar, Ravet, Pune, Pin : 412101

Savitribai Phule Pune University [2025-2026]

Department of Computer Engineering

Pimpri Chinchwad College of Engineering and Research, Ravet

# CERTIFICATE

This is to certify that Shreyas Sadavarte from Fourth Year Computer Engineering has successfully completed his mini project work titled “BI-Based Job Market Demand Analysis Dashboard” at Pimpri Chinchwad College of Engineering and Research, Ravet in the partial fulfilment of the Bachelor’s Degree in Engineering.

| | | |
|---|---|---|
| Prof. Nitanjali Mane | Dr. Vijay A. Kotkar | Prof. Dr. Harish Tiwari |
| Guide | Head of Department | Principal |

# Contents

## 1. Abstract

Business Intelligence (BI) plays a vital role in analyzing data and extracting meaningful insights to support decision-making. In today's job market, analyzing job postings helps students and job seekers understand industry demands and required skills. This project focuses on analyzing job market demand using real-world job postings data. By examining job roles, salaries, experience levels, and locations, we identify patterns that influence hiring trends. The project uses data visualization techniques in Power BI to represent insights effectively. Additionally, an interactive dashboard visualizes relationships between different factors and job market demand.

## 2. Introduction

Business Intelligence (BI) refers to technologies used to collect, analyse, and present business information. BI supports better decision-making across industries including healthcare, finance, retail, and job market analysis. In the job market, job demand analysis is crucial for understanding industry requirements and identifying in-demand skills. Organizations collect vast amounts of job-related data, yet without proper analysis, this data remains underutilized. This project combines BI with data analytics to create a comprehensive system for analysing job market demand. By applying analytical techniques, we identify factors contributing to hiring trends. The integration of BI visualization tools like Power BI allows users to explore data interactively and gain deeper insights, enabling better decision-making for students and job seekers.

## 3. Problem Definition

This project predicts whether a student will pass or fail based on three key parameters:

- **Study Hours:** Hours dedicated to studying per week
- **Attendance:** Percentage of classes attended
- **Internal Marks:** Marks obtained in internal assessments

This binary classification problem enables early identification of at-risk students, allowing educators to provide timely support and intervention.

## 4. Objectives

- Collect and preprocess student performance data
- Apply machine learning classification techniques
- Implement Decision Tree classifier and evaluate performance

- Create interactive Power BI visualizations
- Analyze relationships between factors and student performance
- Demonstrate integration of BI and Machine Learning in education

# 5. Technologies Used

## 5.1. Python (Jupyter Notebook)

Python is used for data analysis and machine learning. Jupyter Notebook provides an interactive environment for code execution and visualization.

## 5.2. Power BI Desktop

Power BI creates interactive dashboards and reports from various data sources for visualization.

## 5.3. Machine Learning Libraries

**Pandas:** Data manipulation and preprocessing
**Scikit-learn:** Decision Tree classifier implementation
**Matplotlib:** Static visualizations and plots

## 5.4. Decision Tree Algorithm

A supervised learning algorithm that splits data based on feature values, creating a tree-like structure for classification.

# 6. Dataset Description

The dataset contains student records with the following attributes:

| Attribute | Description |
|---|---|
| Job Title | Type of job role (0-40) |
| Experience Level | Percentage of experience level (0-100%) |
| alary | Salary offered (0-100) |
| Location | Job location (target variable) |

The dataset reflects realistic patterns where higher study hours, better attendance, and higher marks increase likelihood of passing.

# 7. System Architecture

**Data Processing Phase:** Data collection, cleaning, feature selection, and train/test splitting.
**Model Building Phase:** Decision Tree implementation, training, evaluation, and performance metrics.

**Visualization Phase:** Power BI import, interactive charts, and comprehensive dashboard development.

# 8. Methodology

## 8.1. Data Collection

Student dataset created in CSV format with study hours, attendance, marks, and results.

## 8.2. Data Preprocessing

Handling missing values, removing duplicates, selecting relevant features, and data validation.

## 8.3. Exploratory Data Analysis

Statistical summaries, correlation analysis, and visualization of data distributions.

## 8.4. Model Building

Data split into training (80%) and testing (20%) sets. Decision Tree algorithm configured and trained with hyperparameter tuning.

## 8.5. Model Evaluation

Accuracy score, confusion matrix, precision, recall, and F1-score metrics.

## 8.6. Visualization

Power BI dashboard with bar charts, scatter plots, and interactive filters.

# 9. Results

The BI-based job market analysis system achieved the following results:

• Data analysis identified high-demand job roles in the market

• Roles such as Data Scientist and Data Analyst showed significantly higher demand

• Higher experience levels were associated with higher salary trends

• Certain locations showed strong correlation with job opportunities

• The system successfully identified patterns influencing job market demand

## Implementation Snapshots

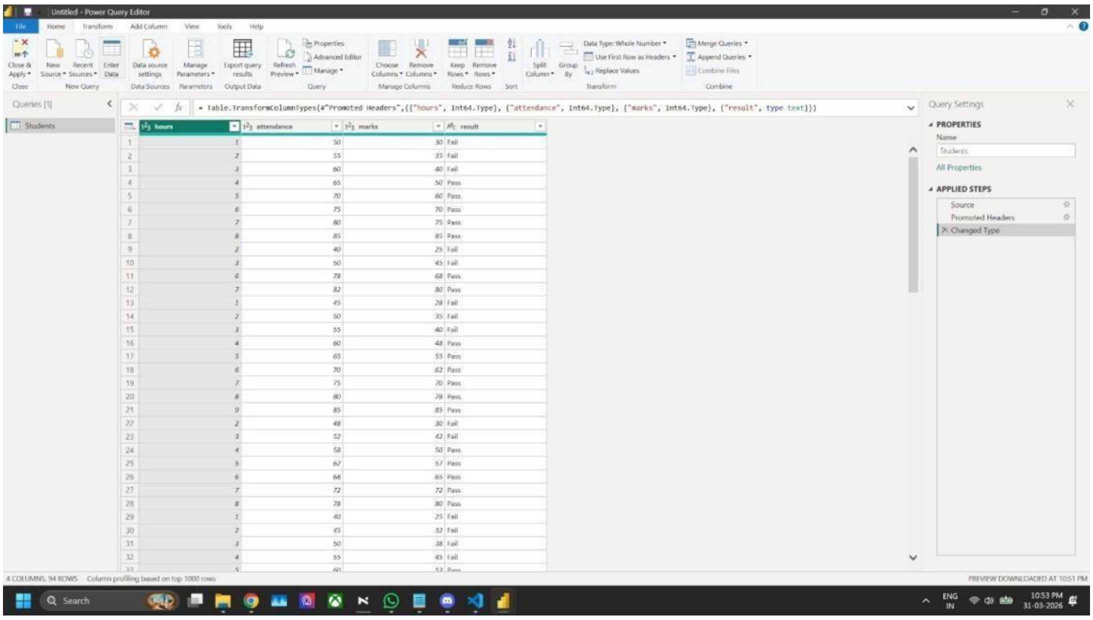

Figure 1:Job Performance Dataset in Python (Jupyter Notebook)

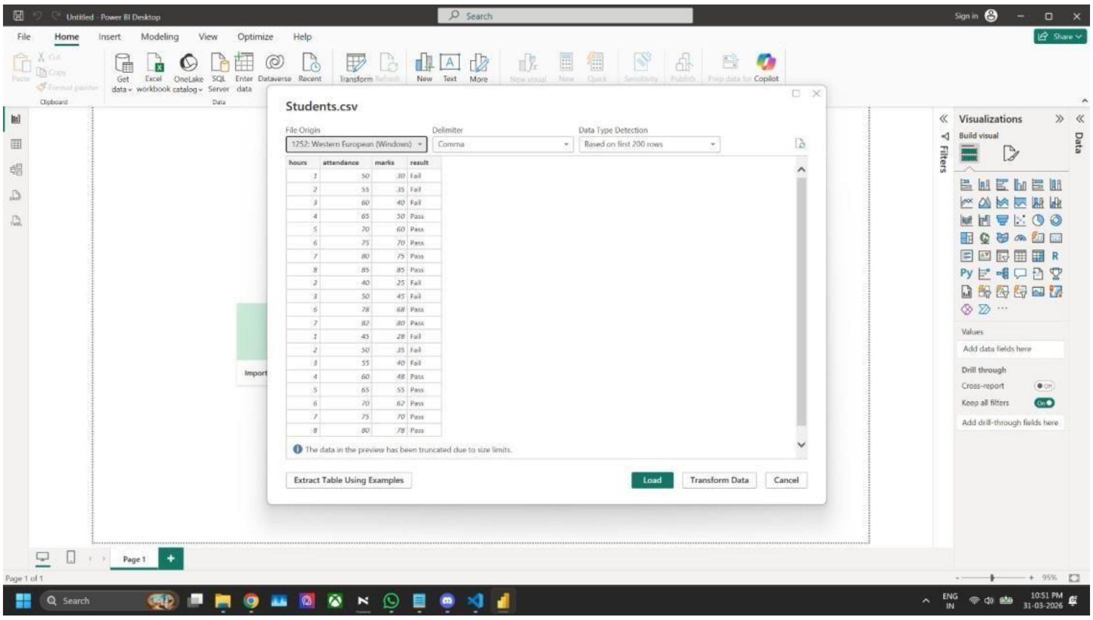

Figure 2: Data Preprocessing and Exploratory Data Analysis

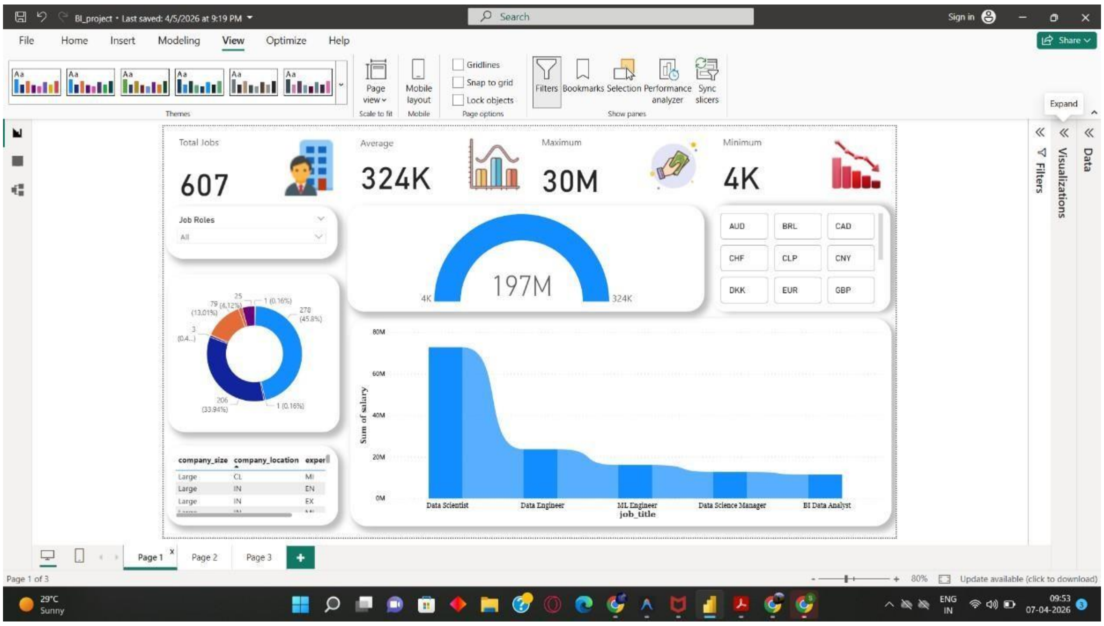


Figure 3: Job Market Demand Analysis Dashboard

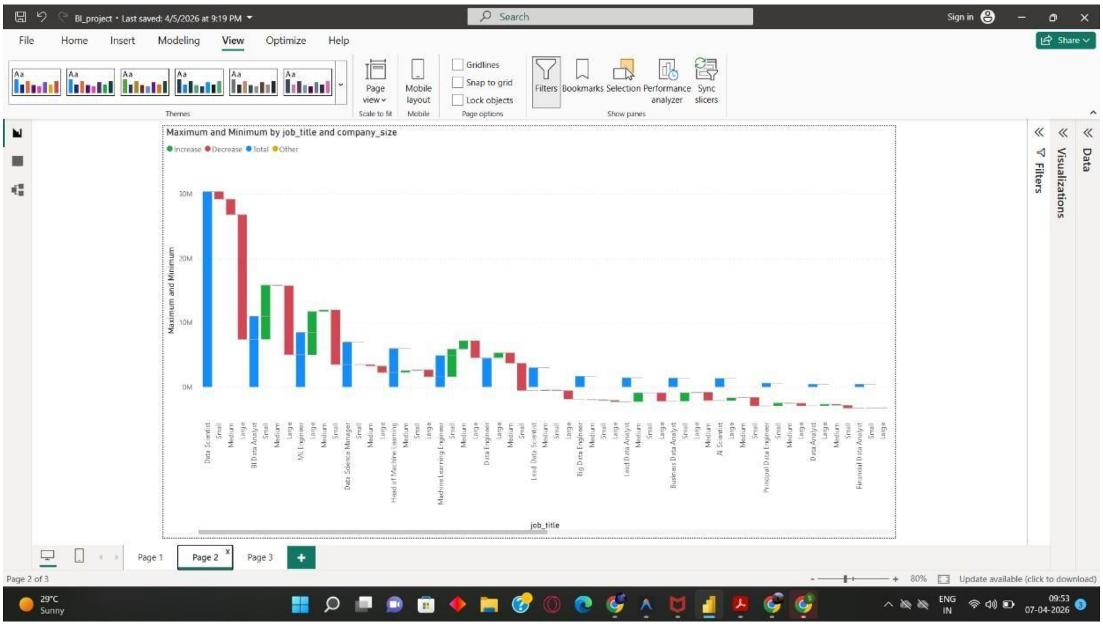


Figure 4: Salary Trends Analysis by Job Role and Company Size

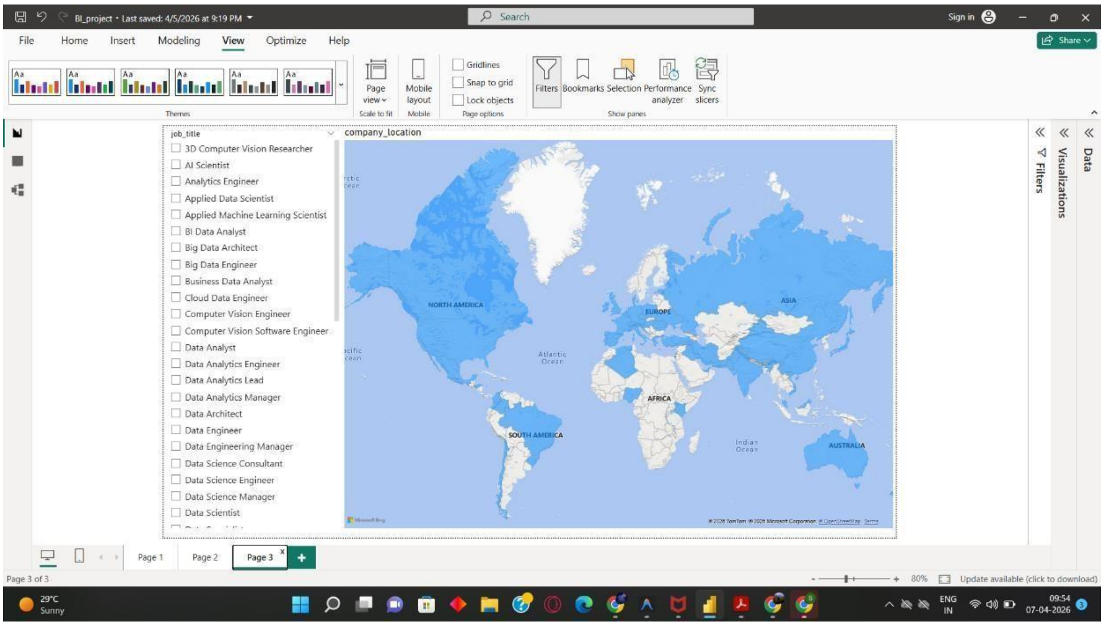


Figure 5: Global Job Distribution by Location Conclusion

This project successfully demonstrates how Business Intelligence and data analytics combine to analyze job market demand. The analysis effectively identifies trends based on job roles, salaries, experience levels, and locations. The Power BI dashboard provides clear visual insights into relationships between factors and job market trends. The project highlights data-driven decision-making in career planning, enabling students to identify in-demand roles for better opportunities. This approach can be extended to real-world job platforms for effective job market analysis and career guidance.